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TITLE OF THE INVENTION

MACHINE LEARNING-BASED FRAUD DETECTION FOR CLOUD BANKING USING BEHAVIORAL BIOMETRICS

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MACHINE LEARNING-BASED FRAUD DETECTION FOR CLOUD BANKING USING BEHAVIORAL BIOMETRICS

Technical Field

[0001] The embodiments herein generally relate to a method for machine learning-based fraud detection for cloud banking using behavioral biometrics.

Description of the Related Art

[0002] The Cloud banking has become widely used as a result of the financial sector's rapid digital transition, allowing institutions to provide consumers in a variety of geographic locations with safe, scalable, and highly available services. Real-time transactions, data-driven decision-making, and easy access to financial products are all made possible by cloud-based platforms. But the attack surface for bad actors has also increased as a result of this interconnected ecosystem.

[0003] Passwords, security tokens, and even static biometric identifiers like fingerprints are examples of traditional authentication methods that are becoming more and more susceptible to sophisticated cyberattacks, phishing scams, credential stuffing, and synthetic identity theft. More dynamic, ongoing, and context-aware security measures are now required as a result of this. One method that has shown promise in tackling these security issues is behavioral biometrics. Behavioral biometrics, as opposed to static biometrics, which are based on fixed physiological characteristics, aims to identify users by analyzing distinctive patterns in their interactions with digital systems, such as the dynamics of keystrokes, mouse movements, touchscreen gestures, navigation speed, or even posture when handling devices. Because they depend on a user's cognitive and physical processes, these characteristics are intrinsically more difficult to mimic, which makes them extremely useful for real-time detection of aberrant behavior or unauthorized access. The capabilities of behavioral biometric solutions have been significantly enhanced by the incorporation of machine learning (ML) into fraud detection systems. Over time, machine learning algorithms can pick up on minute changes in user behavior, adjust to new trends, and spot aberrations that could be signs of fraud. Because fraud attempts frequently take place in milliseconds and static rule-based systems are unable to keep up with the constantly changing strategies of cybercriminals, this capacity is essential in cloud banking. Large-scale, high-dimensional behavioral data can be processed by supervised, unsupervised, and deep learning algorithms to identify anomalies that could otherwise go undetected.

[0004] Cloud banking, behavioral biometrics, and machine learning working together has many benefits. Without sacrificing system efficiency, cloud architecture makes it possible to centrally collect, store, and evaluate behavioral data at scale. This allows for ongoing monitoring of millions of transactions. Simultaneously, dynamic updates and deployment of ML-powered behavioral models guarantee that detection systems continue to be robust against emerging and complex fraud tactics. Fraud prevention is becoming more proactive rather than reactive as a result of research and industry acceptance showing a significant movement toward continuous authentication, where users are authenticated during a session rather than simply at login. Additionally, privacy-preserving strategies like differential privacy and federated learning are being used more and more to protect sensitive user data while adhering to strict financial rules. Essentially, a significant advancement in the security of digital finance is represented by the field of machine learning-based fraud detection for cloud banking employing behavioral biometrics.

SUMMARY

[0005] In view of the foregoing, an embodiment herein provides a method for machine learning-based fraud detection for cloud banking using behavioral biometrics. In some embodiments, wherein the financial industry has undergone a change thanks to the quick uptake of cloud-based banking services, which provide safe, scalable, and on-demand access to banking operations. But this technological development has also opened up new avenues for sophisticated fraud efforts, which frequently get past established security measures. Machine learning (ML)-based fraud detection systems combined with behavioral biometrics have become a reliable and flexible way to counter these changing threats. Keystroke dynamics, mouse movements, touchscreen gestures, and navigation patterns are examples of behavioral biometrics that offer a continuous, non-intrusive layer of authentication that is intrinsically harder for hackers to imitate than static credentials. This method uses cloud-based infrastructure to gather and process massive amounts of user interaction data in real-time, guaranteeing scalability and quick reaction times. To find minute variations from predefined user profiles, machine learning algorithms such as supervised classifiers, anomaly detection models, and deep learning architectures are trained using historical behavioral information. Unusual login habits, transactional patterns, and navigational irregularities that can point to account compromise or fraudulent intent can all be identified by these models. Because machine learning is flexible, the system may gradually increase detection accuracy and effectively respond to new fraud techniques without requiring significant manual reconfiguration.

[0006] In some embodiments, whereas the fraud detection system gains from distributed data storage, high processing capacity, and integration with various banking service channels by utilizing cloud computing, which permits thorough risk assessment across platforms. To improve overall security posture while preserving user comfort, the system can also be integrated with encryption and multi-factor authentication. Since the data used for modeling does not require direct access to sensitive personal information, privacy concerns are also addressed by the use of behavioral biometrics in cloud banking fraud detection. This enhances consumer trust and complies with legal compliance obligations. Nonetheless, there are still issues with regulating latency in real-time detection scenarios, guaranteeing low false-positive rates, and safeguarding behavioral data from abuse.

[0007] These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The embodiments herein will be better understood from the following detailed description with reference to the drawings, in which:

[0009] FIG. 1 illustrates a method for machine learning-based fraud detection for cloud banking using behavioral biometrics according to an embodiment herein.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0010] The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

[0011] FIG. 1 illustrates a method for machine learning-based fraud detection for cloud banking using behavioral biometrics according to an embodiment herein. In some embodiments, the first step in developing a machine learning-based fraud detection system for cloud banking that uses behavioral biometrics is to carefully incorporate sophisticated behavioral analytics into a cloud-native security architecture that can process millions of transactions per second with incredibly low latency. Unlike static identifiers like fingerprints or passwords, behavioral biometrics record the distinct patterns in a person's long-term interactions with gadgets and apps. Unlike static credentials, these patterns which include keyboard dynamics, touch pressure variations, swipe speed and curvature, mouse trajectories, scrolling velocity, and reaction timings are intrinsically harder for fraudsters to imitate. This method functions as a continuous, non-intrusive authentication layer in cloud banking, enhancing current identity verification systems without creating additional hassles for users.

[0012] Real-time behavioral data streams from various digital channels, including web portals, mobile banking apps, smart ATMs, and customer support chat interfaces, are gathered, processed, and turned into feature vectors. A hybrid ensemble of machine learning models then analyzes the feature vectors, which are then turned into actionable fraud risk scores. The system functions as a multi-stage pipeline. A rule-based decision engine uses these scores to decide whether to allow, challenge with more verification, or block a session. By implementing microservices in containerized environments managed by Kubernetes, the architecture takes advantage of the elasticity of cloud computing and guarantees horizontal scalability to accommodate spikes in transaction volume during moments of high banking demand. Three fundamental tenets form the basis of the conceptual model: transparency, adaptability, and proactivity. Adaptability enables the system to learn from changing fraud tactics and user behavior changes; transparency guarantees that both customers and compliance officers can comprehend how fraud decisions are made, promoting trust and regulatory alignment; and proactive measures guarantee that possible fraudulent behavior is detected before financial loss occurs.

[0013] In some embodiments, the data collection in this situation needs to be thorough, ongoing, and safe. The first step in the process is to integrate lightweight software modules, or SDKs, into the digital platforms of the bank. These modules may capture high-fidelity behavioral data while having little effect on the functionality of the application. JavaScript agents track mouse movement vectors, scroll acceleration curves, hover times over interface elements, typing rhythms, and click cadences for web-based platforms. Native SDKs in mobile banking apps record swipe trajectories, pinch and zoom movements, accelerometer and gyroscope readings to determine the handling style of the device, and touch pressure information where the hardware permits it. Smart ATM interfaces that are increasingly based on touchscreen technology are able to capture keypad entry timing, withdrawal menu navigation speed, and finger press duration. The data point is associated with a session identity, timestamped to within milliseconds, and enhanced with contextual data such as IP geolocation, device fingerprints, and connection attributes like packet loss rates and network jitter. In order to ensure that raw behavioral sequences cannot be reverse-engineered to reveal a user's identity without authorized access to secure mapping keys kept in a bank-controlled vault, personally identifiable information is immediately stripped or tokenized at the point of capture. After that, data is sent to the cloud ingestion layer via TLS 1.3 and encrypted using AES-256. Several downstream services, including preprocessing pipelines, real-time anomaly detection engines, and long-term behavioral profile storage, can consume the ingestion system in parallel thanks to its use of event streaming platforms like Apache Kafka or AWS Kinesis.

[0014] In some embodiments, the inherent noise in behavioral biometric data stems from changes in network conditions, device form factors, and environmental factors that impact user interaction. Thus, preprocessing is essential to guaranteeing model accuracy. To eliminate recurring events brought on by instrumentation artifacts or UI rendering delays, raw behavioral streams are deduplicated. Time-series interpolation techniques are used to impute missing data segments that result from packet loss or transient disconnections, or they are deleted based on their importance to session continuation. Swipe speed on a 5-inch smartphone screen cannot be easily compared to the same measure on a 12-inch tablet without scaling changes, demonstrating the importance of feature normalization. To keep features constant across devices, similar standardization is used for gesture amplitudes, mouse distances, and keypress timings. To maintain significant variance, this normalizing procedure combines min-max normalization with z-score scaling, adjusted for each type of feature. To eliminate unlikely observations that can skew model training, outlier removal uses statistical thresholds and unsupervised anomaly scoring. Distributed data processing frameworks like Apache Spark

Streaming are used to create the preprocessing pipeline, guaranteeing sub-second latency while handling gigabytes of incoming behavioral data every day. The data is clean, device-normalized, temporally consistent, and prepared for conversion into high-dimensional behavioral feature vectors at the conclusion of preprocessing.

[0015] In some embodiments, the normalized behavioral streams are converted into representations that capture the stability and uniqueness of a user's interactions through feature engineering. A fundamental behavioral profile is provided by static statistical parameters, such as mean, median, variance of dwell time, standard deviation of flight time between keystrokes, and average swipe curvature. While entropy-based features quantify the randomness of user activities, which is frequently higher in fraud efforts where behavior is chaotic, frequency domain transformations, like the Fast Fourier Transform (FFT), are used to detect periodicities or rhythmic trends in interactions. Device handoff patterns moving between a phone and a desktop during a session, latency spikes associated with multitasking, and departures from previous banking app navigation paths are examples of context-dependent features. Clustering algorithms aggregate users with similar interaction patterns to solve the cold-start problem for new customers without a baseline behavioral profile. This enables models to draw probabilistic conclusions based on group norms. For millisecond retrieval during real-time inference, feature storage depends on in-memory data grids such as Redis or Hazelcast.

[0016] In some embodiments, since fraudulent cases are uncommon typically less than 0.5% of all transactions fraud detection in behavioral biometrics is a high-stakes, unbalanced classification challenge. Therefore, in addition to choosing high-performance methods, the training procedure also addresses the skewed distribution of classes. A hybrid ensemble is used, which combines deep learning architectures (LSTM, GRU) for sequential temporal analysis, supervised learners Data imbalance is handled via under-sampling of benign instances in training batches, cost-sensitive loss functions that penalize misclassification of fraud more severely, and the Synthetic Minority Oversampling Technique (SMOTE), which increases the representation of fraudulent cases. To speed up model convergence, training infrastructure makes use of distributed GPU clusters with mixed precision computation. Bayesian optimization frameworks are used to automate hyperparameter optimization, guaranteeing that every model component performs at its best for the particular transaction and behavioral environment of the bank.

[0017] The models are deployed into a low-latency inference environment that is based on serverless or containerized microservices after they have been trained. The same preprocessing and feature engineering procedures used in training are applied to behavioral data from a live session as it is received, guaranteeing feature parity. The inference engine, which uses AWS SageMaker Endpoints or NVIDIA Triton Inference Server, receives these features and returns a fraud likelihood in 50–150 milliseconds. Following that, the decisioning layer follows business rules established by the bank. For instance, a probability score of 0.9 or higher may result in the instant denial of transactions, whilst scores between 0.7 and 0.9 start step-up authentication, such as OTP verification or facial recognition.

[0018] The detection system must adjust in almost real-time as fraud strategies change constantly. Feedback loops are created in which human fraud analysts examine flagged transactions and then return their findings whether fraudulent, benign, or inconclusive to the model's training set. In order to optimize model progress with the least amount of labeling work, active learning algorithms give priority to labeling uncertain cases. In order to prevent complete retraining cycles, online learning techniques enable models to update progressively as new data becomes available. Mechanisms for detecting concept drift track changes in feature distributions over time, signaling when fraud tactics or user behavior norms have evolved to the point where model recalibration is necessary. To cut down on computational cost, these drift detection modules can initiate partial retraining targeted at impacted features.

[0019] Microservices for data ingestion, preprocessing, feature engineering, inference, decision-making, and feedback are orchestrated by the deployment architecture using Kubernetes. Distributed message queues guarantee robustness against transient processing bottlenecks, while horizontal pod autoscaling dynamically adapts resources to transaction traffic. With load balancing supplied by a global traffic manager, global deployment makes advantage of numerous cloud regions to guarantee low-latency service for customers who are spread out globally. In order to preserve availability even in the case of regional failures, disaster recovery solutions include failover systems and multi-region model replication. To cut down on latency and bandwidth expenses, edge computing nodes may first do anomaly screening before forwarding fine-tuned features to central models.

[0020] Each microservice's detection rates, model confidence distributions, latency measurements, and operational health indicators are shown on a real-time monitoring dashboard. Engineers can be informed of performance degradation before it affects consumers by using automated anomaly detection on key operational indicators. The system can compare known compromised devices, high-risk IP ranges, and new fraud campaign signatures with behavioral anomalies thanks to integration with external fraud intelligence services. By combining intelligence-led security operations with predictive modeling, this integration produces a multi-layered defense.

CLAIMS

I/We Claim:

1. A method for machine learning-based fraud detection for cloud banking using behavioral biometrics, wherein the method comprising:

    a machine learning engine set up to identify, categorize, and reduce any security risks, such as fraud, phishing, and illegal access attempts, by analysing transactional data, user behavior patterns, and network activity in real time.

    one or more data collectors configured to acquire behavioral biometric signals from a user device during interactions with a cloud banking application, the behavioral biometric signals comprising at least one of keystroke dynamics, pointer or touch trajectories, device motion, application navigation patterns, session timing, and in-application gesture signatures;

    a feature-extraction module configured to transform the behavioral biometric signals and contextual telemetry into multi-scale temporal-spatial features;

    a trained machine learning (ML) inference engine configured to compute, from the features, a probability of fraud and a risk score for a current session or transaction;

    a decision policy engine configured to determine an authentication or transaction outcome based on the risk score and one or more adaptive thresholds; and

    an orchestration layer configured to trigger responsive controls comprising step-up authentication, session termination, transaction hold, or human analyst review, wherein the system operates within a cloud computing environment integrated with a banking platform.

Dated this, 13th August, 2025.

Signature:

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ABSTRACT

MACHINE LEARNING-BASED FRAUD DETECTION FOR CLOUD BANKING USING BEHAVIORAL BIOMETRICS

The present invention relates to the development of the necessity for strong and flexible fraud detection systems has increased due to the banking industry's rapid digital transformation, particularly in cloud-based financial ecosystems. Behavioral biometrics, including keystroke dynamics, mouse movement patterns, and touchscreen interactions, are used in this study's Machine Learning (ML)-based fraud detection system to spot unusual activity in cloud banking platforms. In order to identify aberrations suggestive of fraudulent access or transactions in real time, the suggested system combines sophisticated machine learning algorithms with ongoing user behavior monitoring. A scalable cloud infrastructure securely collects, preprocesses, and analyses behavioral data, guaranteeing low latency and excellent detection accuracy. This method improves security without sacrificing user experience by providing constant, non-intrusive monitoring, in contrast to standard authentication methods. The potential of behavioral biometrics in conjunction with machine learning (ML) to provide adaptive, robust, and privacy-preserving fraud prevention in next-generation cloud banking systems is highlighted by experimental assessments that show notable gains in detection rates and false-positive reduction.

FIG.1